How to Write Problem Statement:

1. Describe the "ideal":

- ✓ Provide background context first so that problem (and its solution) are easier to understand for the reader.
- ✓ Start by describing how things should work.

For instance, let's say that you work at a major airline and that you've noticed that the way passengers board your planes is an inefficient use of time and resources. In this case, you might begin your problem statement by describing an ideal situation where the boarding system isn't inefficient that the company should shoot for, like this: "The boarding protocols used by ABC Airlines should aim to get each flight's passengers aboard the plane quickly and efficiently so that the plane can take off as soon as possible. The process of boarding should be optimized for time-efficiency but also should be straightforward enough that it can be easily understood by all passengers."

2. Explain the problem:

- ✓ Succinctly summarize the problem you intend to solve - this cuts to the heart of the issue immediately and positions the most important information in the problem statement near the top, where it's most visible.
- ✓ Start your sentence with phrasing like "However, ..." or "Unfortunately, ..." to show that the problem you've identified.

Let's say that you think you've developed a quicker, more efficient system for getting passengers aboard our planes than the typical "back to front" seating system. In this case, you might continue with a few sentences like, "However, ABC Airline's current passenger boarding system is an inefficient use of the company's time and resources. By wasting employee man hours, the current boarding protocols make the company less competitive, and by contributing to a slow boarding process, they create an unfavorable brand image."

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3. Explain why it's a big deal:

- ✓ Try to highlight the impact of problem on the company or organization etc.
- ✓ Be as exact and specific about the "burden" of the problem — try to specify an exact (e.g. number, dollar amount etc.) for that problems.

For our airline example, you might proceed to explain the problem's financial cost like this: "The inefficiency of the current boarding system represents a significant financial burden for the company. On average, the current boarding system wastes roughly four minutes per boarding session, resulting in a total of 20 wasted man-hours per day across all ABC flights. This represents a waste of roughly $400 per day, or $146,000 per year."

4. Back up your assertions:

- ✓ Back up your claims with reasonable evidence.
- ✓ Start making specific claims about how serious your/the problem is.
- ✓ Start supporting your/the statements with evidence. In some cases, this may be from your own research, from data from a related study or project, or even from reputable third-party sources.

Let's reexamine the sentences used in the previous step. They describe the cost of the problem, but don't explain how this cost was found. A more thorough explanation might include this: "...Based on internal performance tracking data,[1] on average, the current boarding system wastes roughly four minutes per boarding session, resulting in a total of 20 wasted man-hours per day across all ABC flights. Terminal personal are paid an average of $20 per hour, so this represents a waste of roughly $400 per day, or $146,000 per year."

5. Propose a solution:

- ✓ When you've explained what the problem is and why it's so important, proceed to explain how you propose to deal with it.
- ✓ As with the initial statement of your problem, your explanation of your solution should be written to be as clear and concise as possible.
- ✓ Stick to big, important, concrete concepts and leave any minor details for later — you'll have plenty of opportunities to get into every minor aspect of your proposed solution in the body of your proposal.

In our airline example, our solution to the problem of inefficient boarding practices is this new system you've discovered, so you should briefly explain the broad strokes of this new system without getting into the minor details. You might say something like, "Using a modified boarding system proposed by Dr. Edward Right of the Kowlard Business Efficiency Institute which has passengers board the plane from the sides in rather than from the back to the front, ABC Airlines can eliminate these four minutes of waste." You might then go on to explain the basic gist of the new system, but you wouldn't use more than a sentence or two to do this, as the "meat" of our analysis will be in the body of the research.

6. Explain the benefits of the solution:

- ✓ Again, now that you've told your readers what should be done about the problem, it's a very good idea to explain why this solution is a good idea.
- ✓ Focus on what is the primarily impact on the solution that you proposed. For example, which expenses it will reduce, which new forms of revenue it will generate, and so on.
- ✓ Can also explain non-tangible benefits, like improved customer satisfaction, but your total explanation shouldn't be too much longer than a few sentences to a paragraph.

In our example, you might briefly describe how our company could conceivably benefit from the money saved with our solution. A few sentences along these lines might work: "ABC Airlines stands to benefit substantially from the adoption of this new boarding program. For instance, the $146,000 in estimated yearly savings can be re-directed to new sources of revenue, such as expanding its selection of flights to high-demand markets. In addition, by being the first American airline to adopt this solution, ABC stands to gain considerable recognition as an industry trend-setter in the areas of value and convenience."

7. Conclude by summarizing the problem and solution:

- ✓ Conclude with a summary of your/the main arguments that allows you/the transition easily into the main body of the research.
- ✓ There's no need to make the conclusion any longer than it needs to be — try to state, in just a few sentences, the basic gist of what you've described in your problem statement and the approach you intend to take in the body of the research.

In our airline example, you might conclude like this: "Optimization of current boarding protocols or adoption of new, more-effective protocols is crucial for the continued competitiveness of the company. In this research, the alternative boarding protocols developed by Dr. Right are analyzed for their feasibility and steps for effective implementation are suggested." This sums up the main point of the problem statement — that the current boarding procedure isn't very good and that this new one is better — and tells the audience what to expect if they continue reading.

8. Do not forget a thesis statement:

- ✓ The thesis statement is a single sentence that summarizes your/the entire argument, boiling it down to its bare essentials.
- ✓ A good thesis statement identifies both the problem and the solution as succinctly and clearly as possible.

Polishing Your Problem Statement:

1. Be concise:

- ✓ If there's one thing to keep in mind when writing problem statements, it's this. Problem statements shouldn't be any longer than they need to be to accomplish their task of laying out the problem and its solution for the reader. No sentence should be wasted. Any sentence that doesn't directly contribute to the problem statement's goals should be removed. Use clear, direct language. Don't get bogged down in minor details — problem statements should deal only with the essentials of your problem and solution. In general, keep your problem statement as short as possible without sacrificing it's in formativeness.

2. Write for your audience:

- ✓ When making a problem statement, it's important to remember that you're writing for someone else, not for yourself. Different audiences will have different sets of knowledge, different reasons for reading, and different attitudes toward your problem, so try to keep your intended audience in mind as you write. You want your problem statement to be as clear and easy for your audience to understand as possible, which means you may need to change your tone, style, and diction from one audience to another. As you write, try to ask yourself questions like:
    - "Who, specifically, am I writing for?"
    - "Why am I addressing this audience?"
    - "Does this audience know all of the same terms and concepts as I do?"
    - "Does this audience share the same attitude as I do towards this problem?"
    - "Why should my audience care about this problem?"

3. Remember the "five Ws":

- ✓ Problem statements should be as informative as possible in as few words as possible, but shouldn't delve into minute details. If you're ever in doubt of what to include in your problem statement, a smart idea is to try to answer the five Ws (who, what, where, when, and why), plus how. Addressing the five Ws gives your reader a good baseline level of knowledge to understand the problem and solution without treading into unnecessary levels of detail.
    - For instance, if you're writing a problem statement to propose a new building development to your local city council, you might address the five Ws by explaining who the development would benefit, what the development would require, where the development should be, when construction should begin, and why the development is ultimately a smart idea for the city.

4. Use a formal voice:

- ✓ Problem statements are almost always used for serious proposals and projects. Because of this, you'll want to use a formal, dignified writing style (the same as the style hopefully used for the body of the document) in the problem statement. Keep your writing clear, plain, and direct. Don't attempt to win your reader over by taking a friendly or casual tone in your problem statement. Don't use humor or jokes. Don't include pointless asides or anecdotes. Don't use slang or colloquialisms. Good problem statements know that they have a job to accomplish and don't waste any time or ink on unnecessary content.